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Adoption, Use and Success of Business Intelligence Systems: A Systematic Literature Review

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I. Abstract

Purpose – The main objective of this systematic review is to analyze the factors associated with the adoption, use, and success of Business Intelligence (BI) systems in organizations. Thus, it aims to identify the main factors influencing the adoption of these technologies, the most used theoretical models in scientific research, the critical success factors, and the main obstacles associated with the implementation of this type of system.

Design/methodology/approach – Before structuring and writing the systematic review, a search was conducted using the Scopus database. The search was performed using a query composed of several relevant terms related to Business Intelligence, Business Analytics, Decision Support Systems, and theoretical models such as TAM, TOE, and DeLone & McLean. This initial search resulted in 211 scientific articles. Subsequently, and after applying inclusion and exclusion criteria, which will be presented and explored later, three main studies were selected to support a more in-depth and detailed analysis.

Findings – The articles and respective studies analyzed demonstrate that factors such as support from organizational management teams, data quality, organizational preparedness, and information integration have a significant impact on the success of BI systems. To this end, the selected articles primarily use the Technology-Organization-Environment (TOE) and DeLone & McLean Information Systems Success Model to analyze the adoption and use of these technologies. Furthermore, several obstacles to BI implementation were identified, such as strong resistance to change, lack of skills, and difficulties in technological integration.

Originality/value – This review synthesizes different perspectives related to the adoption and success of Business Intelligence systems, contributing to a better understanding of the organizational, technological, and human factors that influence these technologies. As already mentioned, this review was supported by 3 articles: “Exploring Risks in the Adoption of Business Intelligence in SMEs Using the TOE Framework”, “Analysing the Impact of a Business Intelligence System and New Conceptualizations of System Use” and “Business Intelligence System Adoption in the Hotel Industry: Identifying Barriers and Actions”.

Keywords – Business Intelligence; Business Analytics; Decision Support Systems; TOE Framework; DeLone & McLean; Technology Adoption; Systematic Literature Review

Paper type – Literature Review

II. Introduction

A. Context and motivation

Business Intelligence (BI) systems have taken on an increasingly important role in modern organizations due to their ability to transform large volumes of data into useful information to support decision-making, both at the strategic and operational levels. With current technological growth and the consequent digital transformation, organizations are generating increasingly massive amounts of data, increasing the need to introduce tools capable of efficiently collecting, processing, analyzing, and visualizing information.

In this context, BI systems emerge as a fundamental technological solution, allowing organizations to improve operational efficiency, optimize business processes, identify market opportunities, and support data-driven decisions. Furthermore, the growing adoption of technologies such as Big Data, Artificial Intelligence (AI), Business Analytics (BA), and cloud computing has further reinforced the importance of BI systems in various sectors, including healthcare, banking, logistics, tourism, education, and industry.

Despite the increasing investment in BI solutions, many organizations continue to face significant challenges, many of which are related to their effective implementation and use. Many studies already conducted indicate that many BI projects fail to achieve the expected results due to a variety of factors, such as poor data quality, lack of organizational preparedness, insufficient analytical skills, resistance to change, or even technological complexity.

Consequently, understanding the factors that influence the adoption, use, and efficiency of BI systems has become an important research topic in the field of Information Systems.

B. Problem Statement

In fact, Business Intelligence systems are widely recognized as strategic tools for improving organizational performance and supporting decision-making; however, there is still no consensus regarding the main determining factors for their adoption and successful use. Existing scientific articles, particularly those we selected to support our systematic review, present different theoretical perspectives, adoption models, and success factors, frequently applied in distinct organizational contexts.

In addition, while some organizations manage to successfully implement BI systems and gain competitive advantages, others face great difficulties in generating value from these technologies. This can be justified by the fact that technological factors alone are not sufficient to explain the success of BI systems, highlighting the importance of other factors and dimensions, such as organizational, environmental, and human dimensions.

Another problem identified in the literature is related to the fragmentation of research related to the adoption and success of BI systems. Many studies analyze only isolated aspects, such as technological acceptance, organizational preparedness, critical success factors, or system quality, without providing an integrated view of the phenomenon.

Thus, we conducted a systematic review with the aim of synthesizing existing knowledge and identifying the main factors that influence the adoption and success of BI systems, the most widely used theoretical models, the critical success factors associated with implementation, the most frequent barriers, and finally, ways to measure the use and success of these technologies in organizations.

C. Research questions

With the aim of answering the identified research problem, this systematic literature review seeks to answer the following research questions:

- Q1: What are the main factors that influence the adoption of Business Intelligence systems?
- Q2: Which theoretical models are most used to study the adoption and success of BI systems?
- Q3: What are the critical success factors (CSFs) in the implementation of BI systems?
- Q4: What are the main barriers and causes of failure associated with BI systems?
- Q5: How is the use and success of BI systems measured in organizations?

D. Paper Structure

This article is organized as follows. The next chapter describes the methodology used in writing this systematic review, including the search strategy, inclusion and exclusion criteria, and the article selection process based on the PRISMA guidelines. The following chapter presents the main results obtained, organized according to the defined research questions. Chapter 4 discusses the main implications of the results for research and for organizations. Finally, Chapter 5 presents the conclusions, not only in terms of limitations, but also possible directions for future research.

III. Methodology

A. Search strategy

This systematic review was conducted with the objective of identifying scientific studies related to the adoption, use, and success of Business Intelligence (BI) systems in organizations.

In the first phase, the Scopus database was used to conduct the research, due to its strong characteristics, such as scientific relevance, comprehensiveness, and quality of the publications presented.

That said, the search strategy was built based on different combinations of keywords related to:

- Business Intelligence;
- Business Analytics;
- technological adoption;
- implementation;
- critical success factors;
- theoretical models of adoption and success of information systems.

The main query used was the following:

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( TITLE-ABS-KEY (
( "Business Intelligence" OR "BI systems" OR
"Business Analytics" )
AND ( adoption OR implementation OR "critical
success factor*" OR success OR failure )
AND ( organization* OR firm* OR enterprise* OR
compan* )
AND ( "TAM" OR "Technology Acceptance Model"
OR "TOE framework"
OR "Technology-Organization-Environment"
OR "DeLone and McLean"
OR "IS Success Model"
OR "Resource-Based View"
OR "RBV" )
) )

OR

( TITLE-ABS-KEY (
"decision support systems"
AND ( adoption OR success )
AND ( "TAM" OR "TOE" OR "DeLone" )
) )
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This initial search resulted in 211 scientific articles found in the Scopus database.

B. Inclusion and exclusion criteria

Next, in order to ensure the relevance and scientific quality of the studies analyzed, inclusion and exclusion criteria were defined.

Inclusion Criteria

That said, articles were included that:

- addressed Business Intelligence, Business Analytics, or Decision Support Systems;
- analyzed factors related to adoption, implementation, use, or success;
- used theoretical models such as TAM, TOE, DeLone & McLean, or RBV;
- presented an organizational context.

Exclusion Criteria

Studies were excluded if they:

- did not have a direct relationship with Business Intelligence;
- were excessively technical;
- did not address organizational, human, or strategic factors;
- consisted of non-scientific reviews, editorials, or incomplete documents.

C. Study selection process / PRISMA

Next, the remainder of the study selection process followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines.

As described earlier, the search conducted in the Scopus database resulted in 211 potentially relevant scientific articles.

Subsequently, an initial assessment was carried out through the analysis of titles, keywords, and abstracts, allowing the exclusion of studies not directly related to the research objectives.

After this first filtering, a comprehensive evaluation of the articles considered most relevant was performed.

Finally, three articles were selected as the most suitable to support this systematic review.

D. Data extraction and analysis

After the final selection of articles, a systematic extraction of the most relevant information from each study was carried out.

For each article analyzed, elements such as the following were considered:

- Year of publication;
- Organizational context;
- Methodology used;
- Theoretical models applied;
- Adoption factors identified;

- Barriers and critical success factors;
- Main conclusions and contributions.

Subsequently, the articles were organized and analyzed according to the previously defined and presented research questions.

Furthermore, a comparison was made between the theoretical models used by the different authors, as well as between the organizational, technological, environmental, and human factors associated with the adoption and success of this type of system.

IV. Results

A. Factors influencing BI adoption

Based on the articles read and the respective studies carried out, it is stated that the adoption of Business Intelligence (BI) systems is influenced by technological, organizational, and environmental factors. Most articles use the Technology-Organization-Environment (TOE) model to analyze them.

On the technological level, factors that stand out as advantages include compatibility with existing infrastructures, data quality, and system complexity. In the article entitled "Exploring Risks in the Adoption of Business Intelligence in SMEs Using the TOE Framework," the idea is shared that organizations tend to adopt BI systems when they perceive clear benefits and compatibility with systems already in use. On the other hand, a revelation of high levels of complexity can hinder this adoption.

Regarding organizational factors, the importance of support from management and their respective teams, organizational preparedness, and the availability of financial and human resources is evident. In this sense, organizations with better tools and instruments and greater management support have a greater chance of the implementation of BI bringing success and efficiency.

Finally, environmental factors also play a relevant role, especially with issues related to competitiveness, collaboration between organizations, and market demands. In one of the selected articles that support this systematic review, this topic is addressed, insofar as, in the hotel sector, the sharing of information between hotels, distribution platforms, and tourism entities is essential to enhance the value of BI systems.

Thus, it is clear that the adoption of BI depends not only on the technology itself, but also on other factors, such as organizational capacity and the external context in which the organization operates.

B. Theoretical models used

The selected articles utilize different theoretical models to analyze the adoption and success of BI systems.

As already mentioned, the most widely used model was the Technology-Organization-Environment (TOE) model, which clarifies technological adoption through three main dimensions: technology, organization, and external environment. The articles by Ana-Marija Stjepić, Niko Ibrahim, and Putu Wuri Handayani used this model to analyze factors that demonstrate adoption, but also possible obstacles associated with the systems.

Another model identified was the DeLone and McLean model, used in the article by Rolando Gonzales and Jonathan Wareham to evaluate the success of Business Intelligence systems through variables such as information quality, system quality, service quality, user satisfaction, and respective usefulness.

The same authors also analyzed the Seddon model, which seeks to explain the success of information systems through the relationship between system use, user satisfaction, and the advantages obtained. The results demonstrated that this model has strong explanatory power in the context of BI systems.

Thus, and confirming this point, research on BI is strongly supported by theoretical models from the field of Information Systems, especially those related to technological adoption and the evaluation of system success.

C. Critical success factors

In addition to evaluating factors that enhance the value of systems, the studies analyzed identify several critical success factors (CSFs) associated with the implementation of BI systems.

One of the most cited factors is the support provided by management teams and/or directors. This is because organizations where management actively participates in the implementation process tend to achieve better results.

Next, data quality also appears as an essential factor. Incomplete, inconsistent, or poorly integrated data significantly reduces the effectiveness of BI systems and compromises decision-making.

Another relevant factor is organizational preparedness, namely in terms of financial resources, technological infrastructure, and employee skills. Thus, better-prepared organizations have a greater capacity to extract value from the implemented systems.

In addition, authors Rolando Gonzales and Jonathan Wareham share that user satisfaction and the respective usefulness are important indicators of the success of BI systems, being strongly influenced by the quality of information and the service provided by the system.

D. Barriers and causes of failure

Despite all the benefits already shared and associated with BI systems, studies identify several obstacles that can compromise the success of their adoption.

One of the main difficulties identified is related to the lack of organizational resources, resources that are included in financial limitations, the absence of adequate infrastructure, and the lack of technical and analytical skills.

On the other hand, resistance to change and lack of trust in data sharing processes also emerge as relevant obstacles. The authors Niko Ibrahim and Putu Wuri Handayan found that many organizations are hesitant to share information due to concerns related to privacy and competitiveness factors.

In addition, data integration problems and low information quality were also identified as factors that hinder the effective implementation of BI.

Finally, it is noteworthy that the complexity of the systems can represent a significant barrier, especially in organizations with less technological experience.

E. Measurement of BI use and success

The studies analyzed use different metrics to evaluate the use and success of BI systems. As mentioned, the DeLone and McLean model evaluates success through certain dimensions such as information quality, system quality, service quality, user satisfaction, and respective usefulness.

Definitely, the results demonstrate that systems capable of providing relevant, reliable, and useful information tend to generate higher levels of satisfaction and use by users.

In addition, the success revealed in BI systems is often associated with improved decision-making, increased operational efficiency, and strengthened organizational competitiveness. The analysis of the selected articles demonstrates that BI systems create value when they allow data to be transformed into useful knowledge to then support strategic decisions.

V. Discussion

A. Main patterns found

The analysis of the selected studies identified several common patterns regarding issues such as adoption, use, and success of BI systems. In this sense, and firstly, it was found that most studies attribute great importance to organizational factors, especially regarding support from the administrative and/or management team, organizational preparedness, and the existence of a data-driven culture.

The most frequently identified pattern among the articles and respective studies was the strong presence of the TOE model as the main theoretical framework for studying and evaluating the adoption of a BI system. This model appears numerous times due to its ability to simultaneously analyze technological, organizational, and environmental factors.

In addition, it was observed that data quality and information integration are transversal points across all studies. The existence of reliable, accessible, and correctly integrated data is imperative to ensure that BI systems produce useful information for their purpose. In this sense, the concept of an ETL pipeline, or even ELT, is addressed, insofar as operations are carried out whose joint concern is the same quality of data and the efficient and consolidated integration of information.

Regarding the success of systems, studies show that this depends not only on the technical implementation of the solution, but also on user satisfaction, usefulness, and the system's ability to generate value for the organization in question.

Finally, it is revealed that many of the challenges associated with this type of system continue to be related to human and organizational factors, rather than to technological limitations.

B. Practical implications

With the elaboration of this systematic review, we seek to present several practical implications for organizations that intend to implement or improve Business Intelligence systems, based on real-world applications from the analysis of studies already carried out on large organizations.

First, the need for the active involvement of the management and administration team throughout the implementation process becomes evident. Strategic support from management facilitates the availability of resources, increasing the alignment between the system's objectives and the organizational strategy.

Another relevant implication, perhaps even one of the most important, is related to the importance of data quality. As observed at the beginning of this systematic review, more and more organizations produce mass data daily. In this sense, and with the aim of neutralizing the existence of non-standardized data (duplicates, without useful information, etc.), organizations must invest in cohesive data management, cleaning, and integration processes, ensuring that the information used by BI systems is, therefore, consistent, up-to-date, and reliable.

On the other hand, employee training and the development of their respective skills are fundamental to increasing the effective use of the systems. This is because many implementation failures do not stem from the technology itself, but from the difficulty users have in interpreting and using the data produced.

In addition, organizations should not fail to consider external factors, such as competitive issues and those related to regulation and collaboration between partners.

Thus, the implementation of BI should be seen not only as a technological project, but as a serious organizational transformation that seeks to bring efficiency and knowledge to the way data is explored and interpreted.

C. Research gaps

Despite the growing number of studies on Business Intelligence, due to the fact that this type of system seeks to be present in more and more organizations, there are still several gaps in the literature.

Initially, one of the main limitations identified was related to the reduced geographical and sectoral diversity of some studies. Many works focus on specific contexts, such as finance, health, or tourism, making it difficult to generalize the results to other sectors.

Another relevant gap is the reduced attention given to human and behavioral factors. Although some studies address aspects such as user satisfaction or resistance to change, there continues to be little research focused on organizational culture, trust in systems, and users' analytical skills.

It is worth noting that, considering the successive evolutions in the field of Machine Learning and AI agents, there is a growing need to study the impact of Artificial Intelligence and advanced analysis technologies on modern BI systems.

Finally, it is expected that future studies and research can explore integrated models that combine different theoretical perspectives, allowing for a more complete understanding of the factors that influence the success of Business Intelligence systems in organizations.

VI. Conclusion

A. Summary

In conclusion, this systematic review aimed to analyze the factors associated with the adoption, use, and success of Business Intelligence systems in organizations. Through the analysis of the selected articles and studies, it was possible to verify that the adoption of BI depends on a wide range of technological, organizational, and environmental factors. Among the most relevant factors are the support from the respective administrative and managerial teams of the organization in question, data quality, organizational preparedness, technological compatibility, and competitive pressure.

Through the analysis of the study results, we concluded that the most widely used theoretical models for studying this topic are the Technology-Organization-Environment (TOE), the Technology Acceptance Model (TAM), and the DeLone & McLean information systems success model, with the TOE model ultimately standing out. It is worth noting that these models allow us to understand both the factors that influence adoption and the elements that determine the success and continued use of BI systems. Furthermore, it was found that the success of BI systems does not depend exclusively on the technology implemented, but also on other factors, such as the ability of organizations to promote a data-driven culture, leaving aside more fixed and rigid mindsets to change, and to develop skills in their employees.

Finally, it was demonstrated that BI systems can significantly contribute to improving decision-making, increasing operational efficiency, and strengthening organizational competitiveness. This is clear, if a strong and cohesive data processing pipeline is practiced, such as ETL or ELT, as discussed, seeking to eliminate any type of incongruity in the dataset.

B. Future research

Regarding future research, it is considered important to deepen the study of the human and cultural factors associated with the adoption of Business Intelligence, since many projects continue to fail due to resistance to change or lack of skills.

One of the most present topics in current technological developments is the inclusion of Machine Learning techniques and AI agents, so it will also be relevant to investigate the impact of these technologies on modern BI systems.

Another possible field of research involves conducting comparative studies between different sectors of activity or even between organizations of different sizes, allowing for a better understanding of how context influences the adoption and success of BI systems.

In conclusion, future studies could develop integrated models that combine different theoretical perspectives, contributing to a more complete understanding of the factors that condition the effective implementation of Business Intelligence systems in organizations.

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